# Design, Fabrication, and Characterization of All-Glass Laser-Written Vapor Cells for Integrated Photonics Applications

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## 3.1 Initial Considerations & Constraints

Bonding glass and crystal presents unique engineering challenges due to the material's non-porous nature, extreme smoothness, and high surface energy. In industrial applications, selecting the right adhesive or process is critical for ensuring optical clarity, structural integrity, and durability against environmental factors.

Some of the most common glass bonding methods are:

- **UV-Curing Adhesives:** These are one-component resins that remain liquid until exposed to specific wavelengths of ultraviolet light. When this adhesives are exposed to UV light, a rapid polymerization process begins. This method offers strong, precise, and invisible joints, since the refractive index of the adhesive can be tuned to match the substrate's refractive index [1].
- **Anodic Bonding:** The substrates are heated to 300°C–500°C while a high voltage ($\approx$ 1 kV) is applied. Ions in the glass migrate, creating a space-charge region that generates a massive electrostatic attraction, resulting in a permanent chemical bond [2].

While this methods can be very useful for many applications, they present some challenges when it comes to MEMs and Vapor Cells manufacturing. Adhesives present in the interface can easily leak into the sealed cavity and potentially contaminate the gas [3]. This can greatly affect the performance of high-precission devices (e.g. atomic clocks). Moreover, some adhesives can heat a lot when a high-power laser interacts with them. This can introduce stress in the glass cell, causing birefringence, thermal expansion and cracks [4]. Additionally, adhesives do not achieve cell hermeticity standards required for some applications [5]. Most adhesive-free methods, such as anodic bonding, require high temperatures [2]. This introduces the need for more sophisticated ovens and increases the risk of creating stress in the glass. Moreover, some coatings used in optical applications can not be heated to such high temperatures [6].

A modern solution that can adress this two problems at once is *Optical Contact Bonding* [7].

## 3.2 Optical Contact Bonding

Optical contact bonding is a glueless process whereby two closely conformal surfaces are joined, being held purely by intermolecular forces [8].

Generally, intermolecular forces are not strong enough to create a robust bond between two substrates. However, if the surfaces to be bound are conformal with nanometric precision, a sufficient surface area is in close enough contact —also known as *intimate contact*— for the intermolecular interactions to have an observable macroscopic effect [9].

In practice, this means that, in order to bond two glass surfaces together (e.g. two disks to create a wafer), they must meet very strict roughness and waviness requirements [8]. In literature, bonding energies equivalent to the bond density within the bulk material ($2.5\,\mathrm{J\,m^{-2}}$) have been achieved [10]. To obtain such strong bonds, special treatments are often applied to the substrates, in order to ensure surfaces are very clean and ready to form intermolecular bonds between them. To do so, many chemical, plasma, pressure and thermal treatments have proven to be effective [8, 11].

Since Optical Contact Bonding does not require any adhesive, and low-temperature methods have been developed during the last decades [12], it is a promising bonding procedure for glass-based atomic sensors [13].

### 3.2.1 Types of Glasses & Quality Requirements

The success of optical contact bonding critically depends on both the material properties of the glass substrates and their surface quality. Different glass types exhibit distinct chemical compositions, thermal characteristics, and mechanical behaviors that directly influence bonding performance. This section examines the most commonly employed glasses in optical contact bonding applications and establishes the stringent quality requirements necessary for achieving robust, void-free bonds.

#### Fused Silica and Quartz

Fused silica ($SiO_2$, amorphous) and quartz ($SiO_2$, crystalline) represent the gold standard for optical contact bonding. According to the comprehensive characterization by Brückner [14], these materials offer several key advantages:

- **High purity and homogeneity:** Ultra-high purity grades ($>99.99\%$ $SiO_2$) minimize contamination and ensure uniform bonding behavior across the interface.

- **Low thermal expansion coefficient:** With a CTE of approximately $0.5 \times 10^{-6}\,K^{-1}$, fused silica exhibits minimal dimensional changes, thereby reducing thermal stress at the interface.
- **Excellent optical transmission:** High transparency from UV to IR wavelengths (200 nm to 2500 nm) [15].
- **Chemical inertness:** Resistance to most acids and organic solvents allows for aggressive cleaning treatments without substrate degradation.
- **High damage threshold:** Superior resistance to laser-induced damage makes these materials suitable for high-power systems [16].

The primary distinction between fused silica and quartz lies in their structural organization: fused silica is amorphous and isotropic, while quartz is crystalline and exhibits optical anisotropy [14]. For most bonding applications, synthetic fused silica is preferred due to its isotropic properties and exceptional surface quality achievable through polishing [11].

### Borosilicate Glasses

Borosilicate glasses, particularly Borofloat 33 (81% $SiO_2$, 13% $B_2O_3$, 4% $Na_2O/K_2O$, 2% $Al_2O_3$), have gained widespread adoption in MEMS fabrication and vacuum cell manufacturing [17, 18]. These glasses offer:

- **Moderate thermal expansion:** CTE of approximately $3.25 \times 10^{-6}\,K^{-1}$ [19], which is compatible with silicon ($2.6 \times 10^{-6}\,K^{-1}$), making them suitable for glass-to-silicon bonding [7].
- **Good chemical resistance:** Adequate resistance to chemical treatments, though less robust than fused silica [17].
- **Cost-effectiveness:** Significantly lower cost compared to fused silica while maintaining sufficient optical quality for many applications.
- **Commercial availability:** Readily available in wafer formats with polished surfaces meeting bonding requirements.

### Other Optical Glasses

Several specialized glass families are employed in niche applications:

- **Aluminosilicate glasses:** Higher mechanical strength and improved thermal shock resistance compared to standard borosilicates [20]. Lower leak rates for some gasses like Helium [21, 22].
- **Low-expansion glasses:** Featuring CTE values near zero, such as Zerodur, are essential for applications requiring extreme dimensional stability (e.g., space telescopes, lithography systems) [23].

**Surface Quality Requirements**

Achieving successful optical contact bonding demands exceptionally high surface quality across multiple metrics [7, 18]. The following parameters are critical:

1. **Surface Roughness:** Root mean square (RMS) roughness must typically be below 0.5 nm, with peak-to-valley roughness less than 2 nm over the bonding area [7]. This ensures sufficient intimate contact for intermolecular forces to dominate.

2. **Flatness and Waviness:** Total thickness variation (TTV) should be maintained below 1 µm for 100 mm diameter wafers. Surface flatness, typically specified as peak-to-valley deviation over the clear aperture, must be better than $\lambda/10$ (approximately 60 nm at 633 nm) for precision optical applications. Longer spatial wavelength deviations (waviness) are particularly detrimental as they cannot be accommodated by surface deformation during bonding [10].

3. **Particulate Contamination:** The bonding environment must maintain cleanroom conditions (Class 100 or better, ISO 5), as even sub-micron particles can create localized voids that propagate across the interface. Particle counts on the surface should be minimized, with particular attention to particles larger than 0.1 µm [24].

### 3.2.2 Chemical Treatment

Chemical substances are utilized to remove contaminants from the glass or to modify its surface state to be more suitable for bonding. These processes are categorized as *Chemical Cleaning* and *Chemical Activation*, respectively.

The most common chemicals used for cleaning include:

- **Detergents and Solvents**
  - **Micro-Soap:** Eliminates heavy organic matter and solid particles by reducing the surface tension of the water. This allows contaminants to detach from the glass surface, making them easier to remove during rinsing.
  - **Acetone:** Effectively dissolves fats, waxes, and photoresists by breaking down organic polymer chains.
  - **Ethanol:** Typically used as the final solvent cleaning step, as it removes residual chemicals and water. Its high volatility ensures it evaporates quickly, leaving the surface ready for subsequent treatments such as acid baths or plasma activation.

- **Stronger Chemicals**
  - **Piranha Solution:** A mixture of Sulfuric Acid ($H_2SO_4$) and Hydrogen Peroxide ($H_2O_2$). It is extremely corrosive and oxidizing, making it the standard for removing stubborn organic residues.

- **RCA-1:** A solution of Ammonium Hydroxide, Hydrogen Peroxide, and water. It removes light organic films and dislodges insoluble particles from the glass surface through electrostatic repulsion.

- **RCA-2:** A mixture of Hydrochloric Acid, Hydrogen Peroxide, and water used to eliminate heavy metals and alkaline contaminants.

In the case of Fused Silica or Quartz glass ($SiO_2$), surface activation consists of increasing the density of hydroxyl groups (-OH) on the surface. This creates silanol groups (Si-OH), which are essential for successful bonding [**?**]. When two surfaces populated with silanol groups are brought into contact, hydrogen bonds are established between them [**?**]. While these bonds do not initially generate a high-strength joint, they serve as the necessary precursor for permanent covalent bonds, which are finalized through thermal treatment, as detailed in the *Thermal Treatment* section. A summary of this process is illustrated in Figure 1. Generally, surface activation is achieved by soaking the glass in RCA-1 or through plasma treatment.

To maximize the efficiency of these chemical treatments, the use of ultrasonic cleaners is highly recommended during the initial cleaning stages. These devices utilize high-frequency sound waves to create microscopic vacuum bubbles in the liquid —a process known as *cavitation*. The implosion of these bubbles near the glass surface provides a mechanical scrubbing action that removes sub-micron particles and contaminants from pores that are otherwise inaccessible to standard chemical soaking.

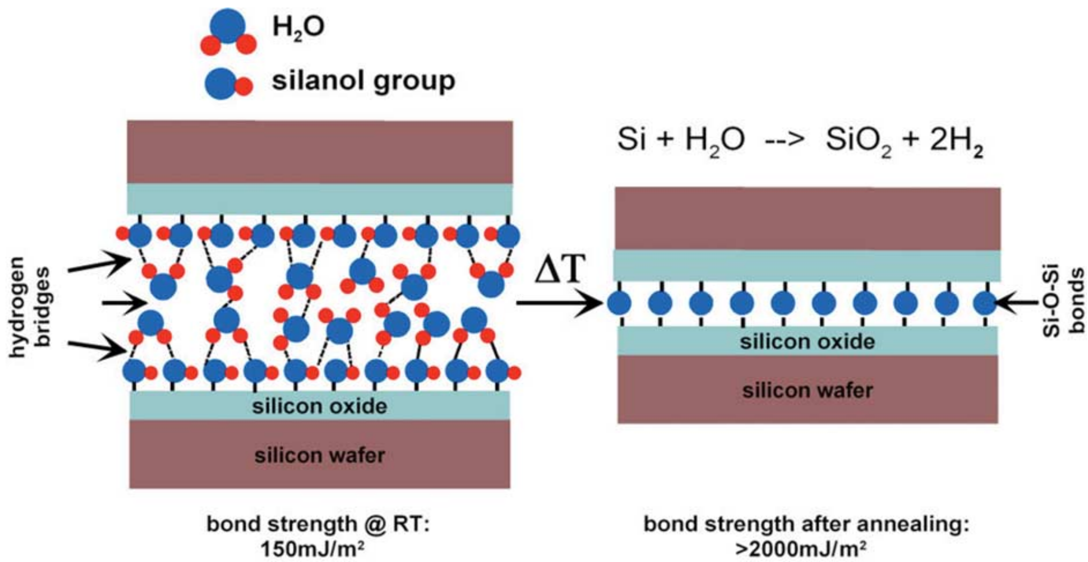


**Figure 1:** Bonding process of two Fused Silica surfaces. Initially, hydrogen bonds are formed between them due to the presence of Silanol (Si-OH) groups. These bonds are subsequently transformed into stronger covalent bonds through thermal treatment.

### 3.2.3 Plasma Treatment

Exposing glass surfaces to plasma —using a plasma cleaner— has proven to be a crucial step to obtain high bonding energies. In the context of glass-to-glass bonding and many other applications, plasma (consisting of electrons, ions, free radicals, and photons) acts both as a cleaning and an activation agent [8, 25, 26]. Figure 2 illustrates some of the many different processes that take place on the surface during plasma treatment; some of the most relevant ones are:

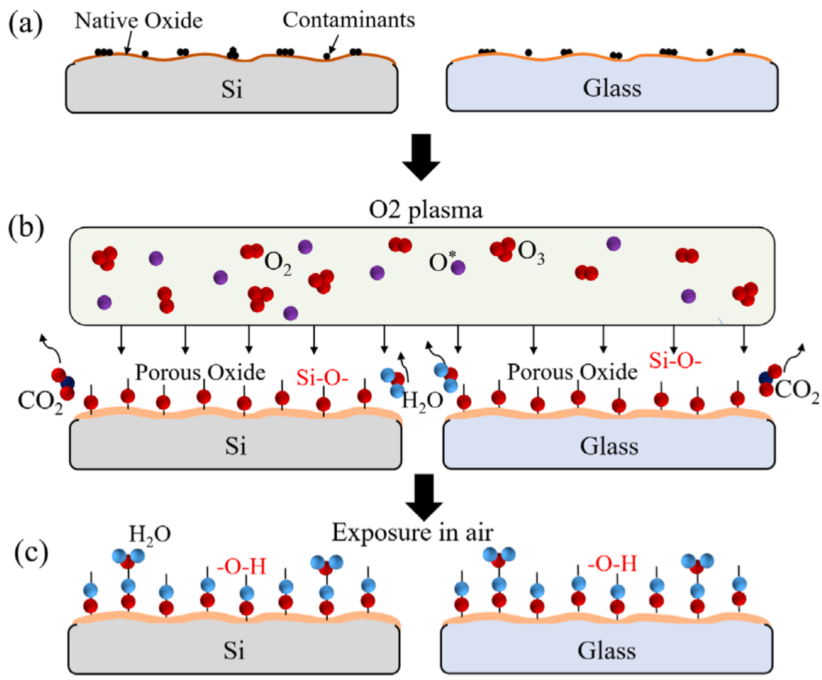


**Figure 2:** Effects of plasma treatment on silica surfaces. Plasma helps to remove contaminants while increasing the number of available silanol groups. Adapted from [27]

1. **Removal of contaminants from the surface:** Plasma removes contaminants from the surface —specially organic matter— and produces more dangling bonds, which leads to higher bonded areas.

2. **Increase in the number of silanol groups:** Plasma initiates a modification of the surface, increasing its number of silanol groups (Si-O-), which are directly correlated to bonding strength.

3. **Improvement of the diffusivity of water and gas trapped at the interface:** Plasma treatment increases diffusivity and thus enhances the evacuation of water and gas trapped between the two glass substrates during the bonding stage. Water evacuation is beneficial as it is needed to replace weak, reversible hydrogen bonds —which play a relevant role to start the bonding process—

by stronger Si-O-Si bonds. Removal of trapped gas is also desirable, since it prevents the formation of bubbles during the annealing step.

Common gases used in these plasma treatments are $O_2$, $N_2$, and $Ar$. When operating a plasma cleaner, special attention must be paid to its power, chamber pressure, gas flow, and operating times, as overexposure can lead to an increase of surface roughness and negatively affect the quality of the final bond.

Exposing plasma-treated surfaces to chemicals should be avoided, since these substances have the capacity to etch or alter the surface, effectively eliminating the benefits of this procedure [28]. Therefore, plasma treatment should be carried out after all chemical cleaning or activation steps. Nevertheless, exposing plasma-treated surfaces to air or deionized water (DIW) has proven to be beneficial, since water molecules can attach to the Si-O- groups present on the surface and create silanol (Si-OH) groups, which are crucial to start the bonding process between two glass surfaces. Therefore, an effective workflow could be chemical cleaning and activation, followed by plasma treatment, a DIW rinse and finally proceeding with pre-bonding and annealing.

Plasma treatment becomes an essential tool in the context of low-temperature optical contact bonding, since it can yield strong bonding energies even at annealing temperatures as low as 100 °C [29].

### 3.2.4 Thermal Treatment

When two sufficiently flat and smooth glass surfaces are brought into contact, the direct bonding process initiates. One of the first mechanisms to occur is the formation of hydrogen bonds. These bonds can subsequently be transformed into stronger covalent bonds, as described by the following chemical reaction:

$$\mathrm{Si-OH + HO-Si \rightleftharpoons Si-O-Si + H_2O}. \tag{0.1}$$

To achieve permanent covalent bonding, water must be removed from the interface, rendering the aforementioned reaction irreversible [8]. This is accomplished through a thermal treatment known as *annealing*.

Annealing involves baking the bonded components in a controlled environment. The primary parameters influencing the outcome are the temperature ramp, the equilibrium temperature, and the annealing time.

The *temperature ramp* refers to the heating rate within the oven. This gradient should be kept as low as possible, as rapid temperature increases can induce thermal stress in the glass substrates. Such stress may compromise mechanical and optical performance, or even result in substrate fracture.

Regarding *equilibrium temperatures*, higher values facilitate greater water diffusivity, thereby removing moisture from the interface more rapidly. Furthermore, increased viscous flow enhances surface contact [8]. However, high temperatures are often unsuitable for precision optics, as they may damage optical coatings or integrated electronics [12]. In such cases, plasma activation is essential, as it increases surface energy and water diffusivity at significantly lower temperatures [30]. This approach is referred to as *Low-Temperature Direct Bonding.*

Concerning *annealing times*, literature indicates that bonding energy increases monotonically over time, whether through a dedicated annealing process or by leaving surfaces in contact at room temperature. The relationship between bonding energy and time follows a logarithmic trend:

$$\gamma \propto \ln\left(\frac{t}{\mu}\right), \tag{0.2}$$

where $t$ is the annealing time and $\mu$ represents a time constant. The role of annealing is effectively to decrease this $\mu$ constant, allowing maximum bonding energy to be reached much faster. Typical annealing times for optical applications range from 2 to 24 hours [30, 27, 12, **?**, 31]. An example of how the bonding energy evolves with time can be seen in Figure 3.

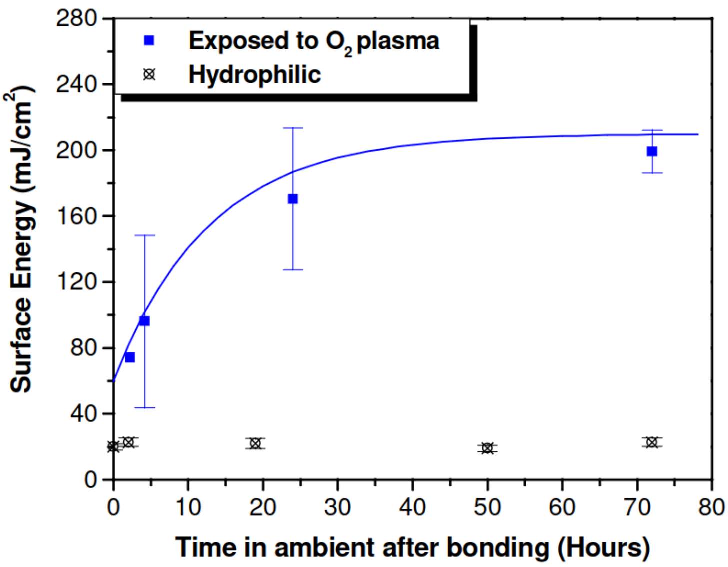


**Figure 3:** Bonding energy evolution with time. From [30]

### 3.2.5 Pressure Treatment

In the context of optical contact bonding, mechanical pressure applied during the bonding phase acts as a catalyst by facilitating the deformation of surface asperities to increase the degree of intimate contact [32, 31], as displayed in Figure 4 This me-

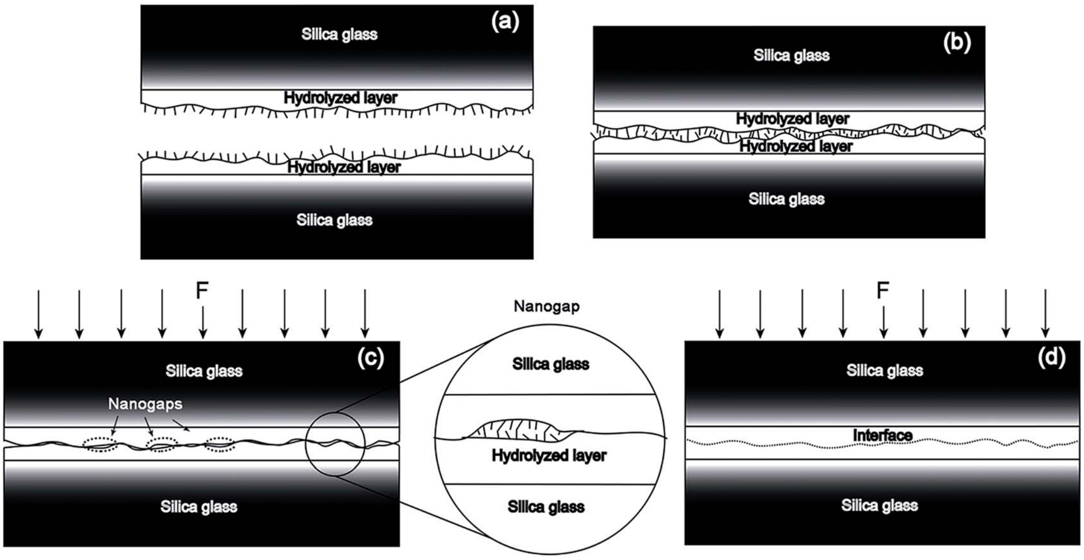


**Figure 4:** Effects of pressure in the bonding process. Interface voids can be prevented by applying forces in the direction perpendicular to the interface. From [32]

chanical assistance accelerates the kinetics of silanol polymerization, allowing for the formation of covalent siloxane bonds at lower temperatures and elevating bond energy from typical room-temperature levels ($<0.5\,\mathrm{J\,m^{-2}}$) to values exceeding $2\,\mathrm{J\,m^{-2}}$, which often surpasses the bulk glass fracture toughness [10].

However, pressure must be carefully optimized; for instance, research on Borofloat 33 wafers indicates that a load of 0.07 MPa maximizes flexural strength, while pressures beyond this limit can lead to stress concentrations and premature failure [31]. Furthermore, non-uniform pressure distributions introduce significant risks to device performance, such as stress-induced birefringence and permanent wavefront distortion [33, 34], necessitating well-planarized pressure plates —with peak-to-valley flatness in the order of micrometers— made of highly stiff materials —such as technical ceramics— [35] for high-precision optical and MEMS applications.

In the setup described by Birckigt et al. [35], pressure is used to enhance contact front propagation during the initial bonding phase. Convex and concave pressure plates —with waviness of 5 µm and 3 µm, respectively— have been employed, as shown in Figure 5. This setup mitigates the risk of creating interface voids and introduces radially symmetric residual stresses.

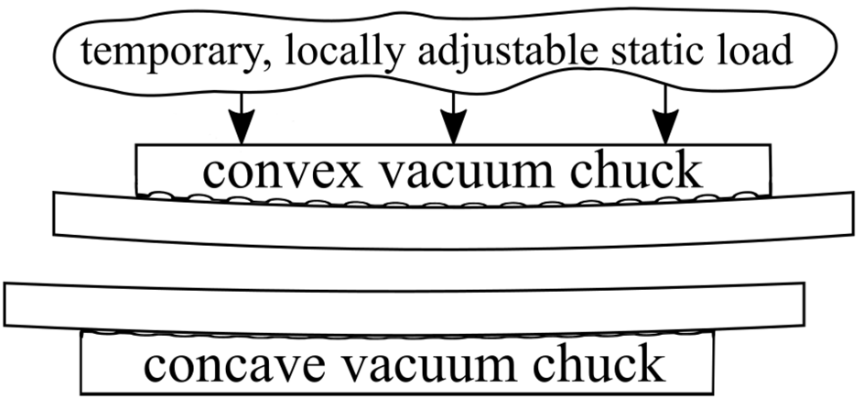


**Figure 5:** Setup described by Birckigt et al. [35]

#### 3.2.6 Step-by-Step Experimental Procedure

### 3.3 Femtosecond Laser Micromachining

While bonding focuses on joining substrates, vapor cells and many applications require the precise removal of glass to create cavities, channels, or complex 3D structures. Due to the transparency, brittleness, and chemical resistance of glass, traditional machining often fails to meet high-precision requirements. **CITA**

- **Mechanical Machining:** Utilizing diamond-tipped tools or ultrasonic drilling. While effective for macro-scale features, it introduces *micro-cracks* and significant internal stress, which can lead to structural failure.
- **Wet Chemical Etching:** Typically involves Hydrofluoric Acid (HF). This process is *isotropic*, meaning it etches in all directions at once. Consequently, it is difficult to create deep, narrow channels with high aspect ratios.
- **Reactive Ion Etching (RIE):** A dry etching process using plasma. It offers high precision but is limited by extremely slow etch rates (micrometers per hour), making it suitable only for surface patterns or thin-film glass.

Femtosecond Laser Writing (FLW) has emerged as a key technology because it allows for "sub-surface" machining without affecting the glass exterior.

- **Nonlinear Absorption:** Glass is transparent to most light; however, a femtosecond laser delivers energy in ultrashort pulses ($10^{-15}$ s). At the focal point, the intensity is high enough to trigger *multiphoton absorption*, allowing the laser to "write" inside the glass without damaging the surface.
- **FLICE Process:** Femtosecond Laser Irradiation followed by Chemical Etching (FLICE) is a two-step technique:
  1. **Irradiation:** The laser modifies the local molecular structure of the glass, increasing its chemical reactivity.

2. **Etching:** The glass is submerged in an etchant (e.g., KOH or HF), which dissolves the laser-modified tracks up to 1000 times faster than the surrounding bulk glass.

- **Zero Heat-Affected Zone (HAZ):** Because the pulse duration is shorter than the thermal diffusion time of the material, the process is considered "cold machining." This prevents melting, burrs, or thermal cracking.

Femtosecond laser technology is uniquely suited for the optics and semiconductor industries because it overcomes the isotropic limitations of chemical etching and the mechanical stresses of traditional drilling. It remains the only viable method for creating "buried" 3D micro-channels and complex internal cavities within transparent substrates. **MILLORAR. POSAR PAPER DE GIANVITO (FOTOS I EXPLICACIONS)**

## 3.4 Post-Treatment

### 3.4.1 Enhanced Helium Hermeticity

Helium plays a critical role in the operation of various atomic sensors, serving primarily as a working medium or buffer gas in high-sensitivity applications (e.g., magnetoencephalography [36]). However, due to its extremely small atomic radius, helium exhibits one of the highest permeation rates through glass walls compared to other gases and alkali metals typically confined within micro-electromechanical systems (MEMS) vapor cells.

For high-precision applications where long-term cell hermeticity is paramount, aluminum oxide ($Al_2O_3$, or alumina) has been identified as a highly effective barrier [21, 22]. Researchers have found that utilizing aluminosilicate glass—a glass matrix enriched with $Al_2O_3$—or depositing thin $Al_2O_3$ coatings directly onto standard glass cells significantly mitigates helium diffusion.

As illustrated in Figure 6, the permeability constant ($K$) of aluminosilicate glass is between two and three orders of magnitude lower than that of conventional borosilicate glass. Furthermore, applying an $Al_2O_3$ coating to borosilicate cells can independently reduce helium permeability by one to two orders of magnitude. Synergistically combining an aluminosilicate glass substrate with the aforementioned $Al_2O_3$ coating yields the best results, achieving ultra-low permeabilities on the order of $10^{-22}$ $m^2s^{-1}Pa^{-1}$.

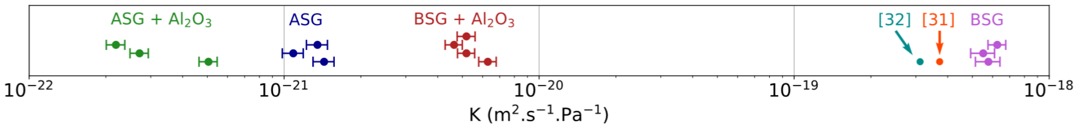


**Figure 6:** Comparison of the permeability constant $K$ across uncoated borofloat, uncoated aluminosilicate, $Al_2O_3$-coated borofloat, and $Al_2O_3$-coated aluminosilicate glasses. From [21].

### 3.4.2 Activation

# 4 Characterization and Quality Control

## 4.1 Visual Inspection

Visual inspection is a primary, non-destructive method for evaluating the integrity of optical contact bonds. The physical principle underlying this technique is thin-film interference, which manifests visibly as Newton's rings when the contacting surfaces are illuminated.

In an ideal optical contact bonding process, the interfacial air gap is completely eliminated. Optically, this results in the disappearance of interference fringes, indicating a defect-free and continuous bond. However, the presence of Newton's rings during or after the bonding process serves as a direct indicator of interfacial defects. These interference patterns occur due to localized variations in the gap thickness between the two surfaces. Issues can be identified and evaluated through this phenomenon are trapped air and voids, particulate contamination and surface flatness deviations [37, 8]. Figure 7 shows an example of Newton rings.

Depending on the optical properties of the materials involved, visual inspection is conducted using different setups. For transparent optical components, monochromatic or white light interferometry is typically used. Conversely, for materials that are opaque in the visible spectrum, such as silicon wafers, infrared (IR) transmission imaging is routinely employed to visualize these interference patterns at the bonding interface [25]. Ultimately, an interfacial region completely devoid of visible Newton's rings serves as the standard criterion for a successful, defect-free optical contact bond.

## 4.2 Bonding Energy Test

The most widely employed technique for measuring wafer bonding strength is the *double cantilever beam* (DCB) method, also known as the blade insertion test, crack opening test, or Maszara's method [38]. It consists of inserting a thin blade or razor (typically 0.1 mm thick) into the bonding interface, which induces crack propagation along the bonded wafer pair [38]. The bonding strength is quantified in terms of the

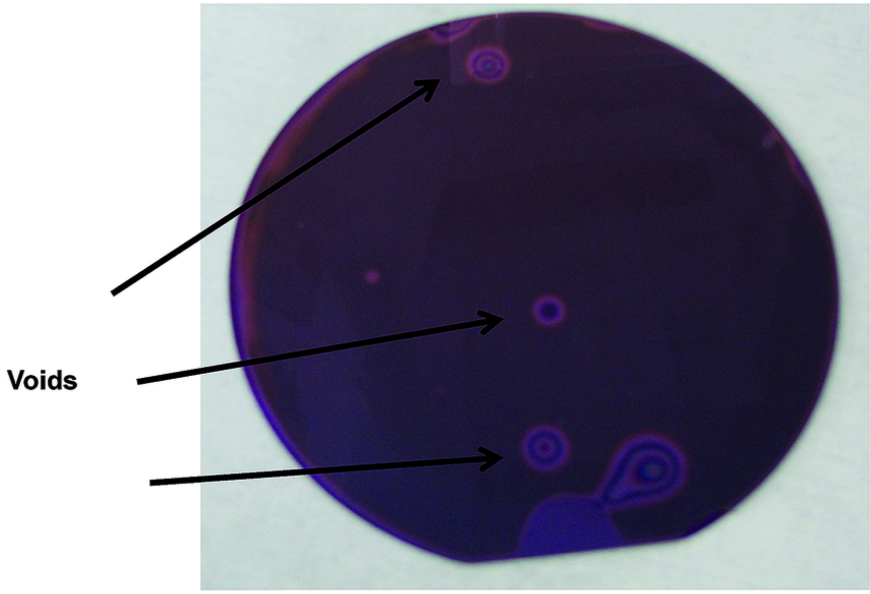


**Figure 7:** Newton rings being formed on the interfase of a SiN-glass wafer. From [37].

energy release rate $G$ (in J m$^{-2}$), which represents the surface energy required to create new surfaces during crack propagation. This parameter can also be expressed as surface energy $\gamma$, where $G = 2\gamma$ for symmetric fracture.

### Theoretical Framework

The energy release rate is calculated based on the energy balance between the elastic strain energy stored in the deformed substrates and the surface energy of the newly created surfaces during crack propagation. For bonded wafers with potentially different materials and thicknesses, the energy release rate is given by [38]:

$$G = \frac{3t^2 E_1 E_2 h_1^3 h_2^3}{8L^2(E_1 h_1^3 + E_2 h_2^3)} \tag{0.3}$$

where $t$ is the blade thickness, $E_1$ and $E_2$ are the Young's moduli of the substrates, $h_1$ and $h_2$ are the substrate thicknesses, and $L$ is the crack length measured from the blade insertion point. For identical substrates ($E_1 = E_2 = E$ and $h_1 = h_2 = h$), this expression simplifies to:

$$G = \frac{3t^2 E h^3}{8L^2} = \frac{3t^2 E}{32y^2} \tag{0.4}$$

where $y = L/(2h)$ is the normalized crack length.

### Experimental Procedure and Considerations

The DCB measurement requires careful attention to several experimental parameters. First, crack length observation should be performed after a sufficient waiting period (typically 210 s) following blade insertion to avoid the effects of static fatigue at the bonding interface [38]. For transparent substrates, crack propagation can be observed optically; for opaque materials such as silicon, infrared imaging or acoustic microscopy is necessary.

### Water Stress Corrosion Effects

A critical consideration in bonding strength measurement, particularly for Si-based materials, is the effect of *water stress corrosion.* This phenomenon occurs when Si–O–Si bonds under applied stress react with water molecules according to the reaction [39]:

$$2\ \mathrm{SiO_2} + \mathrm{H_2O} \rightarrow 2\ \mathrm{Si{-}OH} + \frac{3}{2}\ \mathrm{O_2} \tag{0.5}$$

This water-assisted bond breakage significantly lowers the measured fracture toughness of $SiO_2$ structures, with reductions of up to 50% observed in wet atmospheres compared to dry conditions [39].

### Controlled Atmosphere Measurements

Recent advances have enabled the separation of water stress corrosion effects from ambient water and interfacial residual water through measurements under controlled atmospheres [38]. By performing DCB tests under three distinct conditions—wet atmosphere (ambient air with 50–70% relative humidity), dry atmosphere (pure $N_2$ gas), and vacuum ($\sim 3\times10^{-3}$ Pa)—the individual contributions can be isolated.
Figure 8 illustrates the water stress corrosion mechanism under different measurement atmospheres. In wet atmosphere, ambient water molecules attack the Si–O–Si bonds at the crack tip, reducing the measured bonding strength. In dry atmosphere, only residual interfacial water (trapped during low-temperature bonding) contributes to stress corrosion. Under vacuum conditions, interfacial water is partially desorbed as the crack propagates, minimizing the corrosion effect and yielding the highest measured bond strength [38].

### Advantages and Limitations

The DCB method offers several important advantages: experimental simplicity requiring minimal equipment, direct measurement of surface energy, and applicability to full wafer-level bonding without the need for dicing into smaller specimens [38]. The technique is particularly valuable for process development and quality control in low-temperature bonding applications. However, several limitations must be considered:

- **Measurement range:** The method is limited to bond strengths lower than the intrinsic surface energy of the substrates, typically below 2–3 J $m^{-2}$ for most Si-based materials [38].

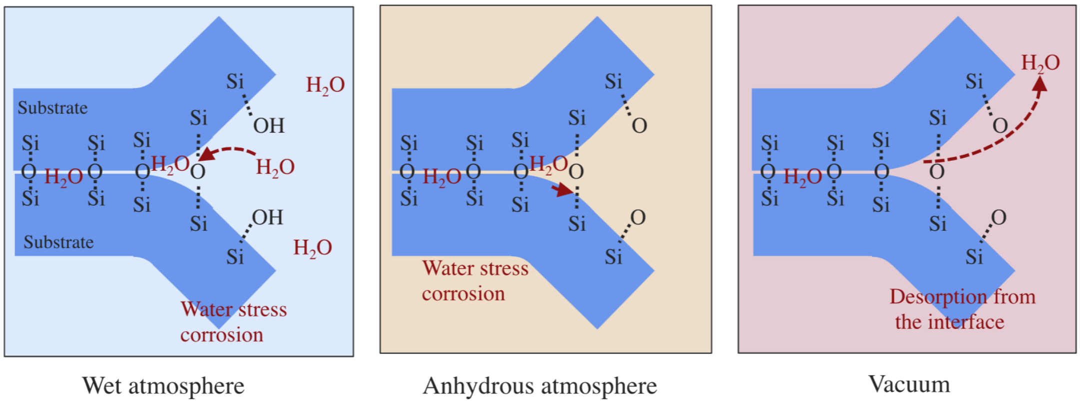


**Figure 8:** Schematic representation of water stress corrosion at the hydrophilic bonding interface during debonding under controlled atmospheres. Left: wet atmosphere showing ambient water attack on Si–O–Si bonds. Center: anhydrous (dry) atmosphere with reduced water availability. Right: vacuum conditions with partial desorption of interfacial water. Adapted from [38].

- **Atmospheric sensitivity:** Results are strongly influenced by measurement atmosphere, with variations up to 50% between wet and dry conditions for $SiO_2$-based interfaces [39].

- **Edge effects and stress concentrations:** Crack initiation at wafer edges can introduce measurement artifacts.

## 4.3 Bonding Strength Test

One of the most widely used methods to assess bonding strength is the *tensile test* (also known as the pull test or direct tensile method). In this approach, bonded substrates are attached to two separate fixtures using a high-strength adhesive. These fixtures are then pulled apart in a direction perpendicular to the bonding interface, applying tensile stress that tends to separate the bonded wafers. A typical experimental setup is shown in Figure 9 [40, 41].

### Fracture Mechanics Analysis

Assuming that the bonding strength is lower than the adhesive's strength, cracks will be produced at the bonding interface when a certain strength threshold is exceeded. The fracture behavior can be characterized using linear elastic fracture mechanics (LEFM). Depending on the crack length, a *stress intensity factor* ($K_I$) can be estimated using the following expression [40, 7]:

$$K_I = \sigma Y \left(\frac{a}{\omega}\right) \sqrt{\pi a} \tag{0.6}$$

where $a$ is the crack length, $\sigma$ is the applied stress, and $\omega$ is the substrate width. The dimensionless parameter $Y\left(\frac{a}{\omega}\right)$ is a geometric correction factor that accounts for the

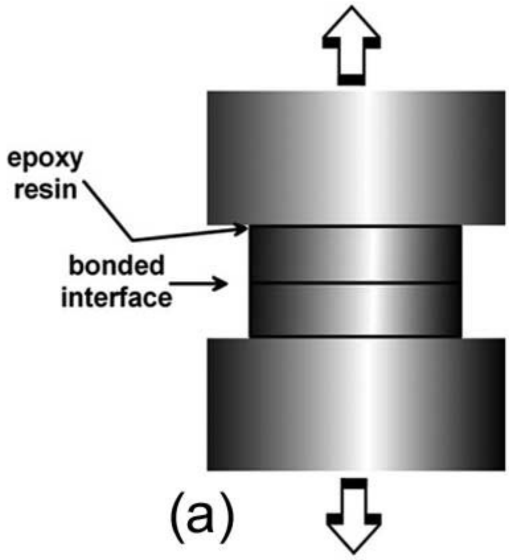


**Figure 9:** Schematic diagram of the tensile test setup for bonding strength measurement. The bonded wafer pair is glued to metal fixtures and subjected to perpendicular tensile force. Adapted from [7].

finite geometry of the wafer and the crack position. For standard edge cracks (the most common failure mode), $Y \approx 1.12$ [42]. For internal cracks located far from the edge, $Y \approx 1.0$ [42].

### Crack Propagation Criterion

Cracks will propagate along the interface —and therefore cause complete fracture— if the stress intensity factor exceeds the material's fracture toughness limit ($K_I > K_{IC}$). This critical value often depends on several parameters, including the crystallographic orientation of the material, temperature, loading rate, and environmental conditions [40, 43].
For brittle materials under mode I loading (tensile opening), the fracture toughness can be theoretically estimated in terms of the surface energy ($\gamma$) as [7]:

$$K_{IC} = \sqrt{\frac{2E\gamma}{(1-\nu^2)}} \tag{0.7}$$

where $E$ is the Young's modulus and $\nu$ is the Poisson ratio of the material.
From Equations 0.6 and 0.7, a critical stress ($\sigma_C$) for crack propagation can be derived:

$$\sigma_C = \sqrt{\frac{2E\gamma}{Y^2\left(\frac{a}{\omega}\right)\pi a(1-\nu^2)}} \tag{0.8}$$

This critical stress represents the bonding strength that can be experimentally measured.

### Advantages and Limitations

The tensile test offers several advantages, including experimental simplicity, direct measurement of bonding strength, and straightforward interpretation based on well-established fracture mechanics principles [41]. However, several limitations must be considered:

- **High measurement variance:** Real experiments exhibit significant scatter in bonding strength values, which can be attributed to local variations in surface parameters such as roughness, total thickness variation (TTV), particle contamination, and bonding defects [41, 7].
- **Edge effects:** Stress concentrations at wafer edges can lead to premature failure that does not accurately represent the average bonding quality [40].
- **Adhesive influence:** The test results may be affected by the properties of the adhesive used to attach the fixtures, particularly if the adhesive fails before the bond [41].
- **Sample preparation:** Careful alignment is crucial to ensure purely tensile loading; any misalignment introduces bending moments that affect the results.

Therefore, while the tensile test provides a useful estimate of bonding strength, results should be interpreted as indicative values rather than absolute measurements. Multiple samples and complementary characterization methods (such as the razor blade test) are recommended for comprehensive bonding quality assessment. **INCLUDE OUR OWN EXPERIMENT HERE**

## 4.4 Surface inspection using an Atomic Force Microscope

*Atomic Force Microscopy* (AFM) is a sophisticated type of scanning probe microscopy, which consists of a physical probe that scans the specimen in order to produce images of surfaces. Atomic Force Microscopes can map the topology of a surface with sub-nanometric resolution, which makes them an ideal tool for surface inspection in the context of atomic sensors, and more specifically in the context of Optical Contact Bonding, due to the strict glass roughness requirements of this method.

The AFM consists of a cantilever with a sharp tip at its end —with a radius of curvature on the order of nanometers, as shown in 10— that is used to scan the specimen surface. When the tip is brought near a sample, forces between them lead to a deflection of the cantilever according to Hooke's law. Forces that are measured in AFM include mechanical contact force, van der Waals forces and capillary forces, among others. Operating modes are divided into static *contact* modes and *tapping* modes, where the cantilever is vibrated at a given frequency.

A brand-new Nanoquartz glass disk, with specified Root Mean Square roughness of

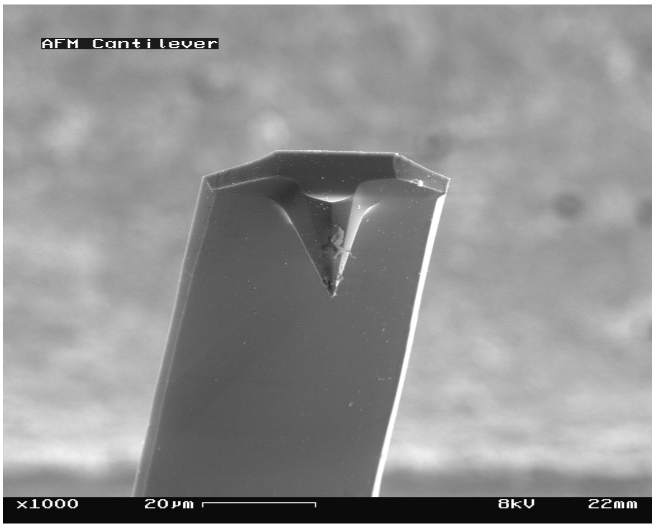


**Figure 10:** View of cantilever in Atomic Force Microscope (magnification 1000x) [44]

500 pm and Total Thickness Variation (TTV) of 2 µm was inspected using a Park NX10 AFM, with a tip Park Scout 350 in *tapping* mode (resonance frequency 320 kHz, rate 0.5 Hz). It was examined in three different spots of its surface. The obtained data was analyzed using the *Gwyddion* software, where two 11 and three-dimensional 12 AFM images of the samples were obtained, along with the density distribution of roughness —displayed in 13. A detailed list of the estimated surface parameters can be found in 1. The precise definition of these parameters can be found in [45].

| **Sample** | $S_q$ **[pm]** | $S_a$ **[pm]** | $S_p$ **[nm]** | $S_v$ **[nm]** | $S_z$ **[nm]** | **Mean height [nm]** |
|---|---|---|---|---|---|---|
| **1** | 518 | 387 | 4.02 | -4.14 | 8.17 | 0.03 |
| **2** | 411 | 298 | 2.04 | 2.83 | 4.87 | 0.02 |
| **3** | 726 | 529 | 7.66 | 4.05 | 11.71 | 0.02 |

**Table 1:** Surface parameters obtained through AFM in three different parts of the surface: *Root mean square roughness* ($S_q$), *Mean roughness* ($S_a$), *Maximum peak height* ($S_p$), *Minimum valley depth* ($S_v$), *Maximum height* ($S_z$) and *Mean height*

The results confirm that the Root Mean Square roughness of the disk is around 500 pm, as suggested by the manufacturer. An alternative estimation of the surface roughness, known as *Mean roughness* (Sa) was found to be in the same order of magnitude. Additionally, it was determined that the total height ($S_q$) is in the range of a few nanometers. The mean height is approximately zero, suggesting that the surface's defects are uniformly distributed. The outstanding surface parameters observed in this disk comply with Optical Contact Bonding requirements and therefore it is suitable for the fabrication of Laser-Written Vapor Cells and high-precission atomic sensors.

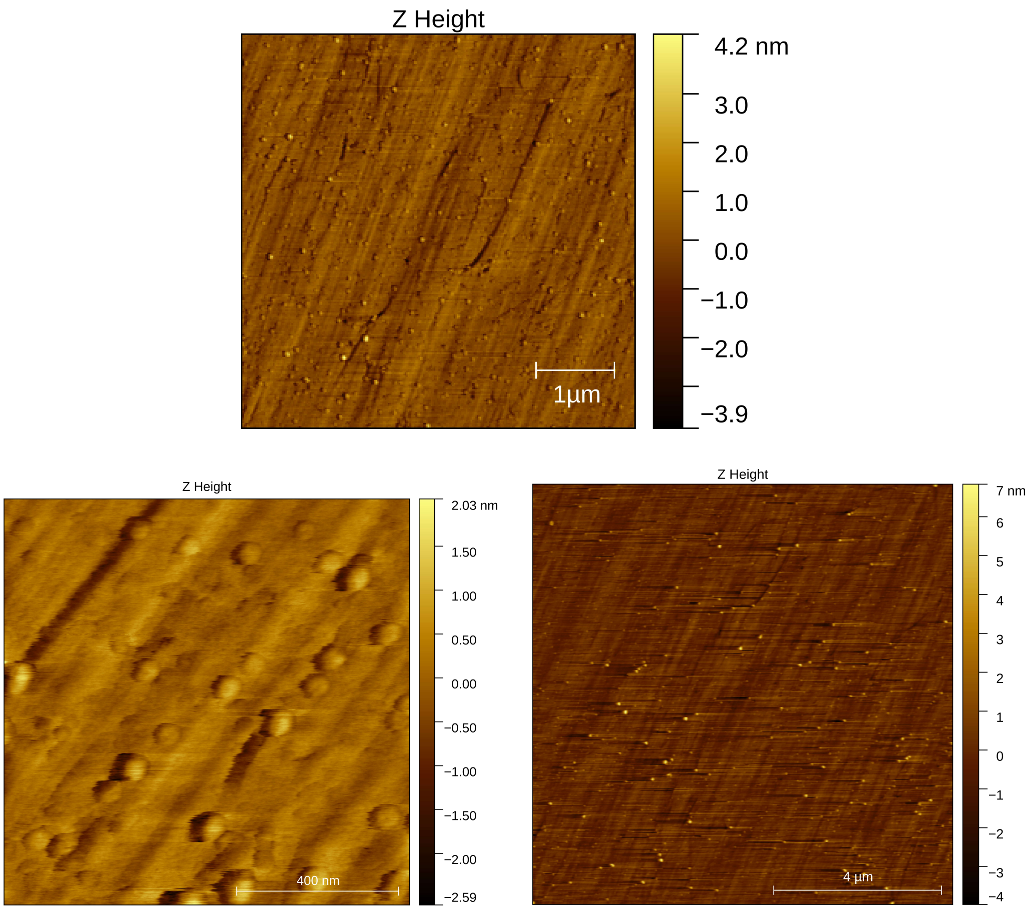


**Figure 11:** 2D Surface images obtained through AFM in three different parts of the pristine Nanoquartz disk's surface.

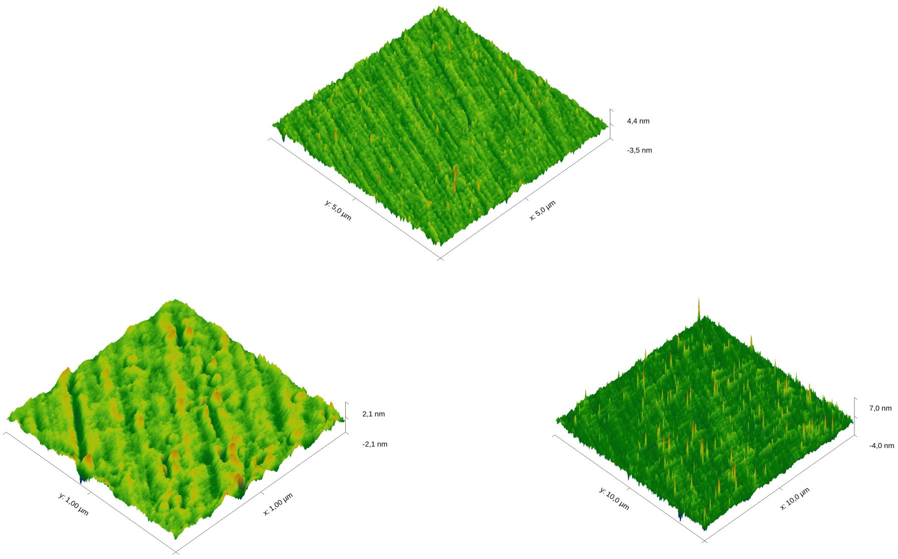


**Figure 12:** 3D Surface images obtained through AFM in three different parts of the pristine Nanoquartz disk's surface.

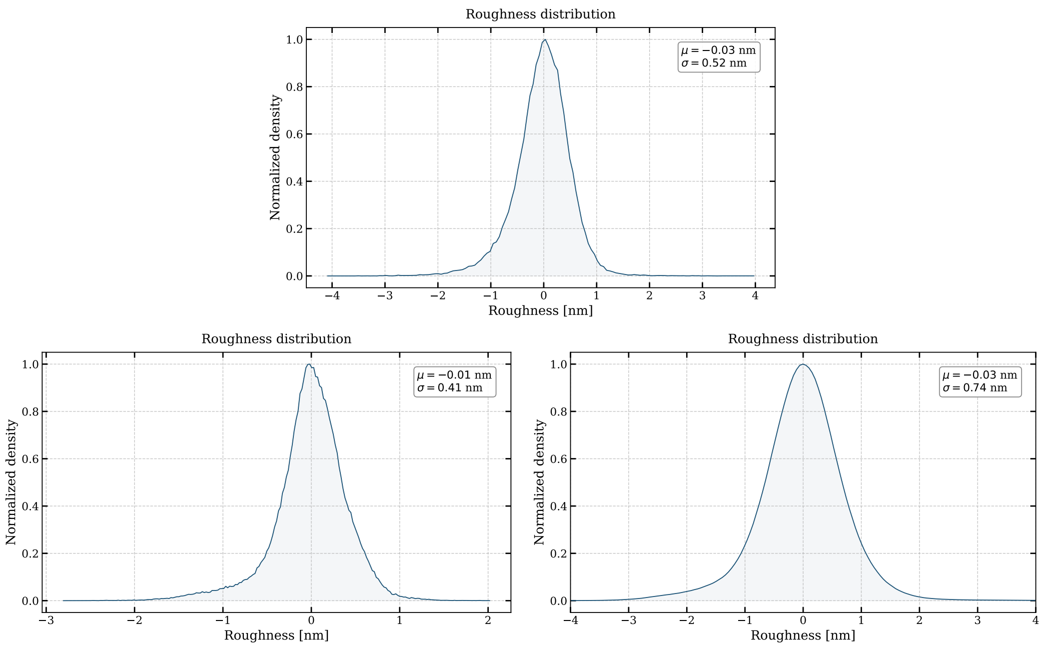


**Figure 13:** Surface roughness distribution obtained through AFM. It can be seen that most of the defects are below the 1 nm range. Furthermore, the mean ($\mu$) is close to zero, confirming that the surface is uniform.

## 4.5 Leak Tests using Spectroscopy

Once the cell is sealed and activated with the alkali metal vapor (typically Rb or Cs), leak tests can be performed over time to estimate its hermeticity. Leak tests involving laser absorption spectroscopy have been reported in literature [46, 47, 48, 49, 50]. In these tests, a tunable laser interacts with the alkali atoms. By measuring the width and central frequency of the resulting absorption profile, the internal environment of the cell can be monitored.

The optical transitions of alkali metals are characterized by the so-called D-line doublet. This fine structure arises from the spin-orbit interaction, which splits the first excited state ($nP$) into two distinct energy levels: $n^2P_{1/2}$ and $n^2P_{3/2}$. The transition from the ground state $n^2S_{1/2}$ to the lower excited state $n^2P_{1/2}$ is known as the D1 line, while the transition to the higher state $n^2P_{3/2}$ is the D2 line. These transitions are illustrated in Figure 14.

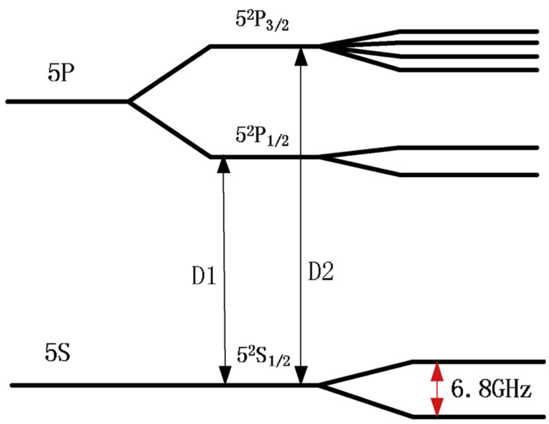


**Figure 14:** Energy level diagram showing the D1 and D2 transitions in an alkali metal atom. From [50]

Collisions between the alkali atoms and other gas molecules perturb the atomic energy levels, leading to collision-induced broadening and a frequency shift. If the cell's hermeticity is compromised and atmospheric gases leak into the cell, the resulting increase in internal pressure ($P$) will cause a detectable width broadening and a frequency shift of the Lorentzian profile. To quantify the gas pressure, the following relations are used:

$$\delta\nu = \delta P \tag{0.9}$$

$$\Delta\nu_L = \gamma P + \gamma_N \tag{0.10}$$

The first equation relates the frequency shift ($\delta\nu$) to the pressure ($P$) through the shift rate constant ($\delta$). The second equation links the pressure to the Lorentzian full width at half maximum ($\Delta\nu_L$), involving the broadening rate constant $\gamma$ and the natural linewidth $\gamma_N$.

Both $\delta$ and $\gamma$ depend on the specific alkali-gas pair and the temperature, with values documented in literature [46, 47, 48, 49, 50]. Furthermore, the broadening rate $\gamma$ at a temperature $T_2$ can be estimated from a known value at $T_1$ using:

$$\gamma(T_2) = \gamma(T_1)\left(\frac{T_1}{T_2}\right)^{\frac{1}{2}} \tag{0.11}$$

Note that the shift constant $\delta$ does not follow such a simple scaling law due to its more complex temperature dependence [46]

### 4.5.1 Experimental Proposal

The spectroscopic framework described above can be directly applied to assess the hermeticity of the fabricated vapor cells. Here, an experimental procedure similar to the one followed by Maurice et al. [51] is described.

First, the sealed and activated cell is heated to a nominal operating temperature $T_{\mathrm{op}}$ —chosen so that the alkali vapor pressure is sufficient to produce a measurable absorption signal— and held there throughout the measurement campaign using a temperature-stabilized mount. Temperature stability is critical, since any drift in $T_{\mathrm{op}}$ would introduce a systematic error in the extracted pressure through the temperature dependence of $\gamma$ and $\delta$ [46].

A tunable laser is scanned across the D1 or D2 absorption line and the transmitted intensity is recorded with a photodetector. To avoid power broadening, the probe beam intensity is kept well below the saturation intensity of the transition. The resulting spectrum is fitted to a Voigt profile, from which the Lorentzian width $\Delta\nu_L$ and the line center $\nu_0$ are extracted. Since the Doppler width depends only on $T_{\mathrm{op}}$ and the atomic mass, the Lorentzian component is unambiguously separated. The internal pressure is then retrieved using Equations 0.9 and 0.10, evaluated at $T_{\mathrm{op}}$ using literature values of $\gamma$ and scaled if necessary via Equation 0.11. The two estimates of $P$ —one from the width and one from the frequency shift— serve as a mutual consistency check.

Spectra are recorded at regular time intervals: initially daily, and subsequently weekly once short-term stability has been confirmed. The pressure $P(t)$ is plotted as a function of time and, in the case of a small but non-zero leak, a linear fit to $P(t)$ yields the leak rate $\dot{P}$. If no detectable increase is observed, the measurement noise propagated through Equation 0.10 sets an upper bound on the leak rate:

$$\sigma_P = \frac{\sigma_{\Delta\nu_L}}{\gamma(T_{\text{op}})} \tag{0.12}$$

where $\sigma_{\Delta\nu_L}$ is the uncertainty on the fitted Lorentzian width. This bound provides a physically meaningful hermeticity statement even in the absence of a detectable leak, and allows direct comparison with benchmarks from the literature: the 707-day vacuum integrity demonstrated by Artusio-Glimpse et al. [13] using Rydberg EIT spectroscopy, and the short-term seal integrity inferred from CPT resonance stability by Kobtsev et al. [52] over timescales up to $\sim$1000 s.

# 5 Experiments & Applications

## 5.1 Rail Tracks Corrosion Inspection

## 5.2 Atomic Clocks

## 5.3 Space Applications

## 5.4 Magnetoencephalography

## 5.5 Magnetometry Experiments on Vapor Cells

In this experiment, the goal was to demonstrate the capacity of vapor cells to serve as precise magnetometers in a laboratory setting. As described in [53, 54], when circularly polarized light interacts with alkali atoms (such as Rubidium or Cesium), it *transfers* some of its angular momentum into the gas. This effectively generates a non-zero electron spin polarization state inside the vapor cell.

In the presence of an external magnetic field $\vec{B}$, the atomic spins undergo Larmor precession at the frequency:

$$\omega_L = \gamma B \tag{0.13}$$

where $\gamma$ is the gyromagnetic ratio ($2\pi$ 3.5 kHz/G for Cesium-133). This precession modulates the projection of the atomic spin along the probe beam direction, causing periodic variations in the optical absorption and thus in the transmitted light intensity detected by the photodiode. When a linear magnetic field ramp —such as $B(t) = B_0 + \alpha t$— is applied across the vapor cell, the transmitted light intensity exhibits a Lorentzian resonance peak centered at $B = 0$. This phenomenon, known as *Hanle resonance*, arises from the following mechanism:

For $B \neq 0$, the Zeeman sublevels ($m_F = -F, -F+1, \ldots, +F$) are non-degenerate and precess at different rates, leading to dephasing of the atomic coherences. This dephasing destroys the spin polarization created by optical pumping, resulting in increased absorption and reduced transmission.

At $B = 0$, all Zeeman sublevels become degenerate, eliminating the precession-induced dephasing. The atomic coherences remain phase-locked, maintaining maximal spin polarization in the direction of the pump beam. Atoms in this polarized state exhibit reduced absorption for the circularly polarized probe light, producing a transmission maximum.

By analyzing the Lorentzian profile, properties of the vapor cell, such as the coherence lifetime ($\tau$), can be determined. The relationship between the Lorentzian full width at half maximum (FWHM) and $\tau$ is given by:

$$\Delta B_{\mathrm{FWHM}} \sim \frac{1}{\gamma\tau} \tag{0.14}$$

### 5.5.1 Experimental Setup

A Twinleaf Vapor Cell, filled with Cesium and Nitrogen as buffer gas, was placed inside a magnetic shield to isolate it from background noise (e.g., Earth's magnetic field). A laser tuned to the D1 transition of Cs —$\lambda$ =895 nm— is directed into the vapor cell. To control the power of the incoming light, a half-wave plate (HWP) and a polarizing beam splitter (PBS) were employed. Just before entering the magnetic shield, a quarter-wave plate (QWP) is placed to generate a circularly polarized beam.

Inside the magnetic shield, coils are used to generate an alternating magnetic field ramp ($B_x = B_0 + \alpha t$) along the beam axis (X-axis). As explained previously, this time-dependent field generates a time-dependent absorption profile, which is detected by a photodetector placed outside the magnetic cavity and connected to a lock-in amplifier. To correct for background noise, DC fields are applied along the Z- and Y-axes. This experiment was carried at room temperature. An illustration of this setup is displayed in Figure 15.

### 5.5.2 Results and Analysis

Measurements were made using different DC field configurations to study their effects on Lorentzian peak height, width, and noise. Figure 16 compares the fitted Lorentzian profiles detected by the photodetector for different DC field configurations.

From these results, it can be deduced that the chosen DC field configuration greatly affects the sharpness of the observed Hanle resonance. Signal amplitudes below 10 mV are obtained for the least optimal configuration ($B_y = 3600$ nT, $B_z = 0$ nT), while peaks greater than 40 mV are achieved with the configuration $B_y = 0$ nT, $B_z = 4815$ nT.

In conclusion, for this specific experimental setup, applying a DC field in the Y direction reduces the amplitude and increases the FWHM of the Hanle resonance, whereas a DC field in the Z direction has the opposite effect.

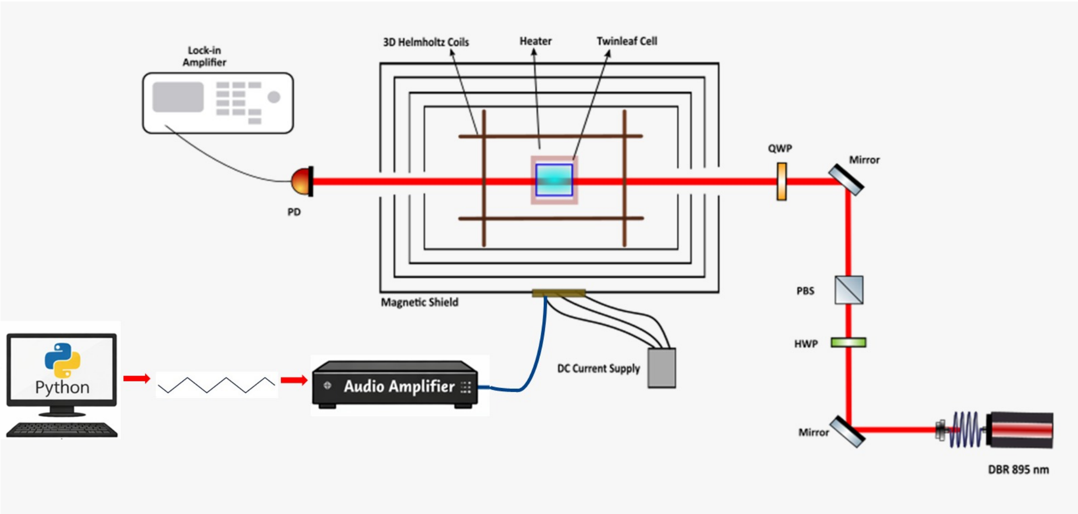


**Figure 15:** Experimental setup used to detect Hanle resonance

Since the goal of this experiment is to study Hanle resonances, the configuration $B_y = 0\,\mathrm{nT}$, $B_z = 4815\,\mathrm{nT}$ was further analyzed. Figure 17 shows the raw data and Lorentzian fit for this configuration. The amplitude of the resonance signal is 42.2 mV. It can be observed that the Lorentzian's maximum is not located exactly at the origin, but is displaced slightly to the right ($B_x = 865.5\,\mathrm{nT}$). This effect is observed systematically and can be attributed to background noise and electronic effects. A FWHM of 8726.3 nT is estimated. From this value, and using 0.14, the relaxation time of this cell at room temperature is ∼50 ms.

# 6 Overlook & Conclusions

# 7 Appendix

## 7.1 Recipes in literature

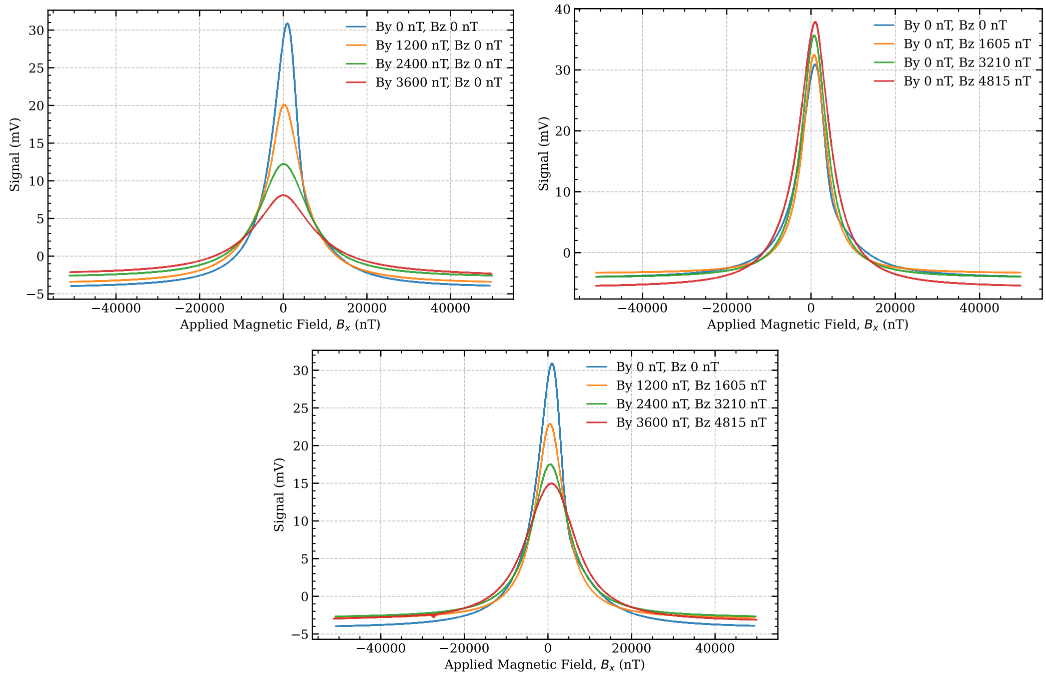


**Figure 16:** Lorentzian profiles when (a) $B_z$ is varied and $B_y = 0$, (b) $B_y$ is varied and $B_z = 0$, and (c) $B_y$ and $B_z$ are varied simultaneously

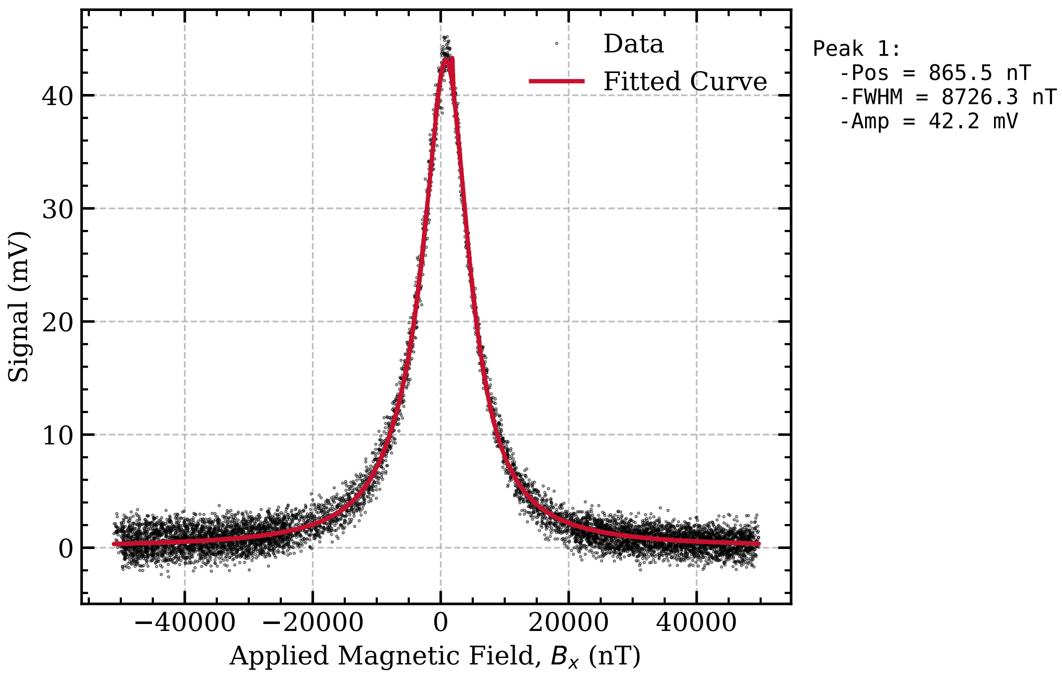


**Figure 17:** Raw data and Lorentzian fit of the Hanle resonance obtained for the optimal DC field configuration ($B_y = 0\,\text{nT}$, $B_z = 4815\,\text{nT}$)

**Table 2:** Summary of chemical cleaning procedures, surface activation methods, and resulting bonding qualities.

| Reference | Chemical Cleaning Steps | | | | | | | Surface Activation | Post-Treatment | Bonding Strength |
|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | | | |
| Domenico Tuli Thesis | DIW 10' 25°C | — | — | — | — | — | — | — | — | 0.06 J/m$^2$ |
| | Acetone 10' 25°C | — | — | — | — | — | — | Plasma 30s 15W 35mTorr | — | 0.06 J/m$^2$ |
| | Acetone 10' 25°C | DIW 10' 25°C | Micro Soap 10' 25°C | DIW 10' 25°C | — | — | — | — | — | 0.06 J/m$^2$ |
| | Acetone 10' 25°C | DIW 10' 25°C | RCA1 10' 60°C | DIW 10' 25°C | — | — | — | — | — | 0.06 J/m$^2$ |
| | Acetone 10' 25°C | DIW 10' 25°C | RCA1 10' 60°C | $NH_4OH$ 2' 25°C | — | — | — | — | — | 0.06 J/m$^2$ |
| | Acetone 10' 25°C | DIW 10' 25°C | Micro Soap 10' 25°C | RCA1 10' 60°C | DIW 10' 25°C | — | — | — | — | 0.06 J/m$^2$ |
| | Acetone 10' 25°C | DIW 10' 25°C | Micro Soap 10' 25°C | RCA1 10' 60°C | DIW 10' 25°C | — | — | — | — | 0.1 J/m$^2$ |
| | Acetone 10' 25°C | DIW 10' 25°C | Micro Soap 10' 25°C | RCA1 10' 60°C | $NH_4OH$ 2' 25°C | — | — | — | — | 0.1 J/m$^2$ |
| | Acetone 10' 25°C | DIW 10' 25°C | Piranha 10' 60°C | DIW 10' 25°C | — | — | — | Plasma 30s 15W 35mTorr | — | 0.1 J/m$^2$ |
| | Acetone 10' 25°C | DIW 10' 25°C | Piranha 10' 60°C | DIW 10' 25°C | Micro Soap 10' 25°C | RCA1 10' 60°C | DIW 10' 25°C | — | — | 0.1 J/m$^2$ |
| | Acetone 10' 25°C | DIW 10' 25°C | Piranha 10' 60°C | DIW 10' 25°C | Micro Soap 10' 25°C | RCA1 10' 60°C | $NH_4OH$ 2' 25°C | — | — | 0.1 J/m$^2$ |
| Cho et al. 2001 | HF (dipped) | Piranha 15' | RCA1 15' | — | — | — | — | $O_2$ Plasma 15-120s 50-300W | Annealing 50h >105°C | 2.5 J/m$^2$ |
| Eichler et al. 2008 | RCA1 | RCA2 | — | — | — | — | — | $O_2$ Plasma 40s 30W 1atm | Annealing 6h 120°C | 2.2 J/m$^2$ |
| Li et al. 2013 | SC1 10' 80°C | SC2 10' 80s | DIW (rinse) | RCA1 10' 60°C | — | — | — | $O_2$ Plasma 60s 50W 20mTorr | Press. 3x10$^5$ Pa → Annealing 6h 200-800°C | 11 MPa BS |
| Wang et al. 2020 | Acetone | Ethanol | DIW | — | — | — | — | $O_2$ Plasma 90s 200W 375mTorr | Wait 12h Room T → Annealing 12h 150°C | 5.5 MPa BS |
| Takei et al. 2010 | RCA1 10' 70°C | DIW (rinse) | HF 2% | — | — | — | — | $O_2$ Plasma 30s 400W 900mTorr | Annealing 8h 250°C 5MPa | 1 MPa BS |
| Lee et al. 2022 | — | — | — | — | — | — | — | $N_2$ Plasma 15s 100W 100mTorr | DIW → Annealing 1-4h 200-500°C | 2.5x w/o plasma |
| Suni et al. 2002 | — | — | — | — | — | — | — | $N_2$ Plasma 30s 50-150W | RCA1 → DIW 5' → Annealing → Dicing → Annealing | >2.5 J/m$^2$ |
| | — | — | — | — | — | — | — | $O_2$ Plasma 45s 50-150W | Annealing 2h 100°C → Dicing → Annealing 2h 200°C | ~2.5 J/m$^2$ |
| Park et al. 2025 | Piranha | HF dip | — | — | — | — | — | $O_2$ Plasma 30s 800W 800mTorr | Press. 5' 5000mBar → Annealing 1h 400°C | ~1.7 J/m$^2$ |
| Our proc. (P11-P13) | Acetone 10' 25°C | DIW 5' + dry | Piranha 10' 75°C | DIW 5' + dry | RCA1 10' 75°C | DIW 5' + dry | — | $O_2$ Plasma 60s 30W 1200mTorr<br>$O_2$ Plasma 120s 30W 1200mTorr<br>$O_2$ Plasma 180s 30W 1200mTorr | DIW 5' → Annealing >12h 150°C | TBD |
| Our proc. (P21-P23) | Acetone 10' 25°C | DIW 5' + dry | Piranha 10' 75°C | DIW 5' + dry | — | — | — | $O_2$ Plasma 60s 30W 1200mTorr<br>$O_2$ Plasma 120s 30W 1200mTorr<br>$O_2$ Plasma 180s 30W 1200mTorr | DIW 5' → Annealing >12h 150°C | TBD |

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